Money, inflation and activity

How does a policy rate connect spending today to payments tomorrow?

Short run and long run

Short run	Long-run benchmark
Some prices and wages adjust slowly.	Prices and wages have adjusted.
Demand can move output away from potential: $Y \neq Y^*$.	The output gap closes: $Y = Y^*$.
Policy can change real borrowing costs and activity.	Real fundamentals shape the real interest rate.

Purely nominal shocks mainly affect prices in the long-run benchmark. The horizon has no fixed calendar cutoff. Real shocks can move potential output itself.

Money, production and spending



A loan can finance a new oven. Interest rates affect the investment decision.

Today's route

Rates, purchasing power and the long-run real benchmark.

Money demand and classic IS–LM with a fixed money supply.

Inflation feedback, then modern rate targeting at the end.

LECTURE 2

Interest and inflation

What is the price of spending earlier?

A zero-coupon bond exchanges money across time

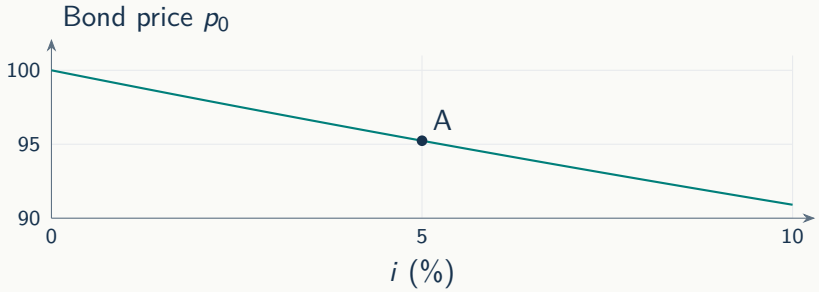
A zero-coupon bond makes one payment at maturity.

$$p_0 = \frac{F}{(1+i)^T}, \qquad F = 100.$$

p_0: price today. i: annual nominal yield. T: years to maturity.

$$T = 1, \quad i = 5\% \quad \Rightarrow \quad p_0 = \frac{100}{1.05} = 95.24.$$

A higher yield means a lower bond price



$$\frac{\partial p_0}{\partial i} = -\frac{TF}{(1+i)^{T+1}} < 0.$$

With face value 100, a one-year price falls from 95.24 at 5% to 90.91 at 10%.

Three rate symbols are enough

Symbol	Meaning
i	Nominal interest rate
$\bar{i}$	Central bank's chosen policy-rate target
r	Real interest rate, measured in purchasing power

Each rate has a stated horizon. In the classic IS–LM model, i is a market outcome.

A real return measures purchasing power

Nominal return counts money. Real return counts what it buys.

$$1 + r_{t+1}^{\text{realised}} = \frac{1 + i_t}{1 + \pi_{t+1}}.$$

Exact purchasing-power conversion: return and realised inflation cover the same period.

The Fisher approximation comes from logarithms

$$\ln(1+r) = \ln(1+i) - \ln(1+\pi).$$

For small rates, $\ln(1+x) \approx x$.

$$\boxed{r \approx i - \pi.}$$

Exact in log gross rates; approximate in simple percentage rates. This line uses realised inflation.

For decisions today, expected inflation matters

$$\boxed{r_t \approx i_t - \pi_t^e, \qquad \pi_t^e \equiv \mathbb{E}_t \pi_{t+1}.}$$

π_t^e: expected inflation over the investment horizon.

$$i_t = 5\%, \quad \pi_t^e = 3\% \quad \Rightarrow \quad r_t \approx 2\%.$$

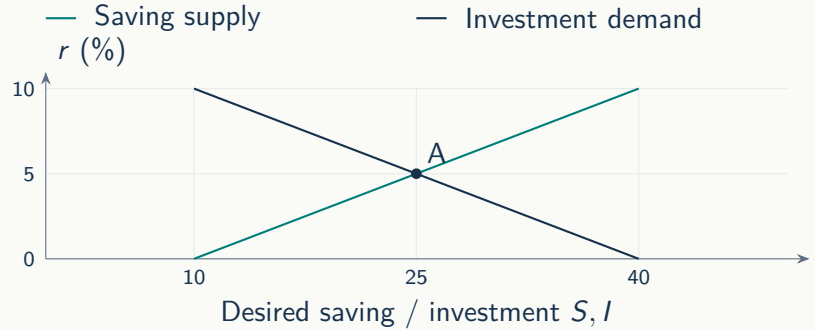
From here, r denotes the approximate expected real rate relevant for decisions.

A nominal cut need not reduce the real rate

	Before	After
Nominal rate i	5%	4%
Expected inflation	3%	1%
Approximate real rate r	2%	3%

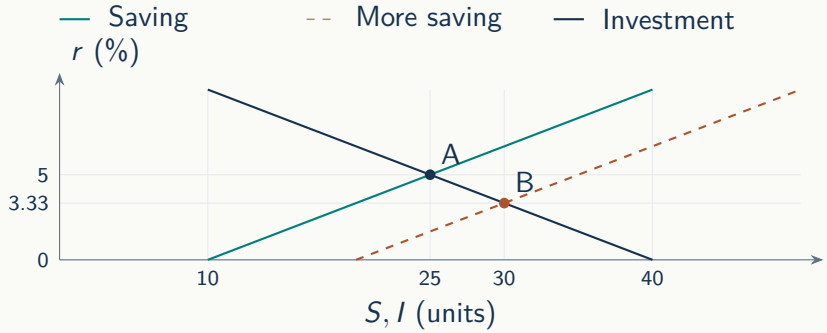
Expected inflation fell faster than the nominal rate.

Saving, investment and the real rate



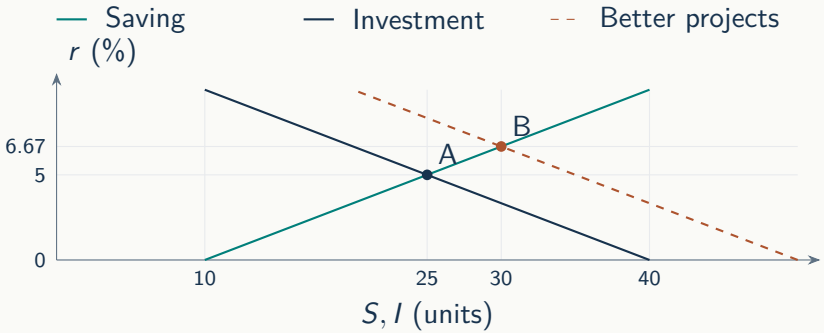
At A: planned saving and investment both equal 25; r = 5%. Hypothetical units.

A larger harvest can lower the real rate



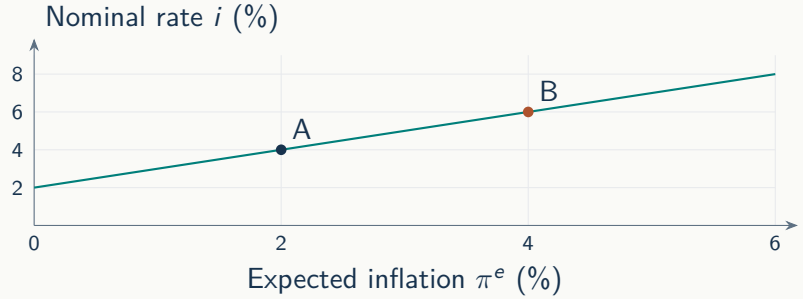
Extra income raises saving supply by 10. The real rate falls from 5% to 3.33%.

Better projects can raise the real rate



Firms want 10 more units at each rate. Competition for resources raises r from 5% to 6.67%.

The long-run Fisher effect



$$i \approx r^* + \pi^e, \qquad r^* = 2\% \text{ held fixed.}$$

With r* fixed, expected inflation and the nominal rate both rise by about 2 points.

The causal story depends on the horizon

Long-run benchmark	Short-run experiment
Resources, tastes and technology pin down the real benchmark. $r = r^*, \quad i \approx r^* + \pi^e$.	Sticky prices and expectations can adjust slowly. Policy changes $\bar{i}$; then $r \approx \bar{i} - \pi^e$ changes.

A higher nominal rate can reflect higher expected inflation or tighter real restraint. Specify the experiment.

Inflation measures a change in the price level

P_t: the cost index of a representative basket.

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

Inflation means the same money buys less of that basket.

Where does the extra money for higher prices come from? The next example needs none.

Prices can rise without any extra money

One-good economy, one period	Before	Supply shortfall
Money spent in the period	100	100
Goods available	10	5
Price per good	10	20

The same spending competes for fewer goods.

Fixed nominal spending and no extra money in this hypothetical example.

New money and faster spending are other possibilities

$$MV = PY, \qquad V \equiv \frac{PY}{M}.$$

Bank lending can create new deposits.

Existing balances can circulate faster: the same money supports more spending.

M is a money stock; PY is nominal output over a period; V is its turnover ratio.

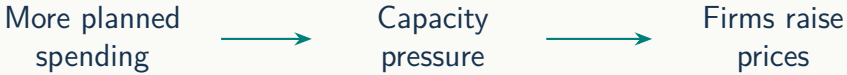
Disinflation can leave prices higher

Date	Basket price	Inflation
0	100	–
1	110	10%
2	112.20	2%

Inflation fell. The price level continued to rise.

Demand pressure can raise prices

Orders grow faster than firms can expand production.



The effect depends on how much productive slack remains.

Supply shocks can raise prices while output weakens

An energy shortage makes production more expensive.

Firms may raise prices and produce less at the same time.

A central bank can influence demand. It cannot create the missing energy.

Expectations can make inflation persistent

Expected cost increases enter wage negotiations and pricing plans.

Contracts can carry an initial shock into later periods.

Wage catch-up alone does not establish an explosive wage-price spiral.

Why high or volatile inflation is costly

Unindexed income and savings lose purchasing power.

Unexpected inflation redistributes value in nominal contracts.

Unstable prices make planning and relative-price signals harder.

Why choose a small positive target?

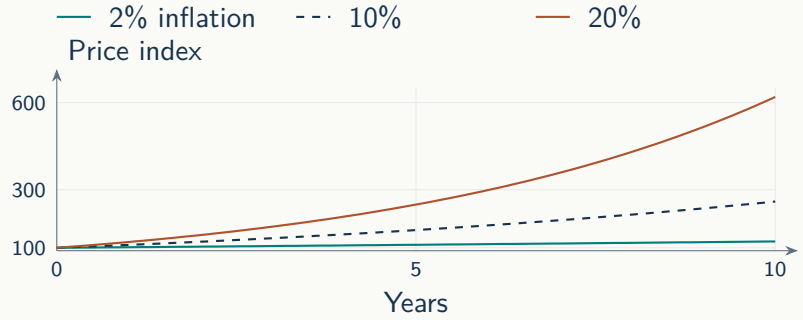
A buffer against persistent deflation.

More room to lower real rates when nominal rates face a lower bound.

Easier real-wage adjustment when nominal wages are sticky.

The ECB targets 2% inflation over the medium term. This is a policy choice.

Constant inflation compounds the price level



$$P_t = P_0(1+\pi)^t.$$

LECTURE 2

Money and short rates

What can a central bank actually change?

A bank loan creates an asset and a deposit

Bank assets	Bank liabilities
Loan to borrower: +100	Borrower's deposit: +100

The borrower receives spendable money and an equal debt.

Lending still faces credit, capital, funding and liquidity constraints.

The central bank anchors short-term market rates

Banks use central-bank reserves to settle payments.

Administered rates and operations implement the overnight target $\bar{i}$.

Long yields also depend on expected future policy and risk premia. Exceptional regimes target them directly.

Exception: a central bank can target a long yield

In 2016, the BoJ introduced yield curve control.

It bought Japanese government bonds to keep the ten-year yield around 0%.

$$p_0 = \frac{100}{(1+i)^T}: \quad \text{support the bond price to restrain its yield.}$$

A historical policy example. The ZCB formula illustrates the price/yield mechanism.

Money is useful, but costly to hold

Money settles purchases and provides liquidity.

Holding it can give up interest on an alternative asset.

Real money balances measure purchasing power

$$\text{Real balances} = \frac{M}{P}.$$

A representative basket	Before	Prices double
Nominal money M	100	100
Price P per basket	10	20
Baskets the balance can buy	10	5

M/P is purchasing power held for transactions, measured as a stock.

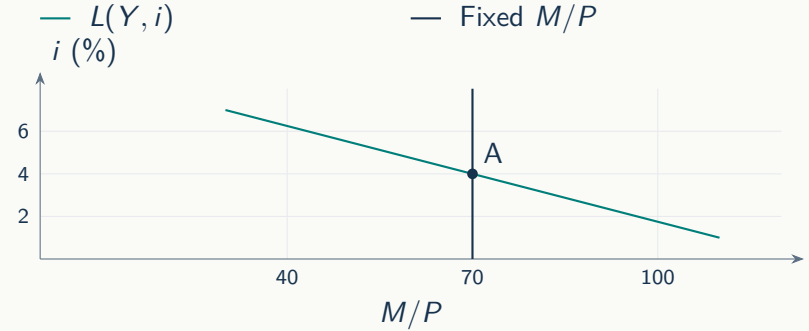
The money-market condition

$$\frac{M}{P} = L(Y, i), \qquad L_Y > 0, \quad L_i < 0.$$

Symbol	Meaning
M	Nominal money stock chosen in the classic model
P	Price level, fixed in this experiment
L(Y,i)	Desired real money holdings

More real transactions raise money demand. Higher bond interest makes cash more costly to hold.

The money-demand diagram



Classic money market: fixed prices P and real output Y. Hypothetical units.

LECTURE 2

IS–LM and shocks

What changes after spending or policy changes?

What is an economic model?

Like a model in physics: isolate a mechanism and state what is fixed.

For now, we care about slopes, derivatives, intersections and shifts.

Specific numbers and functional forms help draw examples; they are assumptions.

The aim here is economic intuition. Quantitative forecasts later require measurement and estimation.

The variables and assumptions of this model

Variable	Meaning
Y	Real output and the income generated
i	Nominal market short rate
P	Price level, fixed here

Expected inflation and credit spreads are fixed. Compare short-run equilibria, possibly below potential.

A firm needs 10 million to start a project

Today: 8 million borrowed + 2 million of the owner's funds.

One year later: the project pays 15 million.

$$\text{Debt repayment} = 8(1 + i).$$

The owner compares the project with an alternative

$$\text{Equity payoff} = 15 - 8(1 + i).$$

$$\text{Alternative payoff} = 2(1 + i).$$

Cash left after repaying debt must also compensate the owner for the alternative investment.

The project's net present value

$$NPV = \frac{15 - 8(1+i)}{1+i} - 2$$

$$\boxed{NPV = \frac{15}{1+i} - 10.}$$

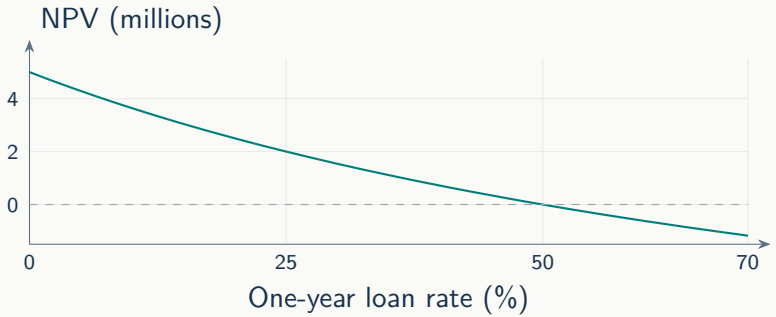
NPV is value today after subtracting today's investment.
Both expressions use the same certainty and discount-rate assumptions.

Working through the financing cost

Loan rate	Debt repayment	Equity payoff	Project NPV
5%	8.40	6.60	4.29
25%	10.00	5.00	2.00
50%	12.00	3.00	0.00
60%	12.80	2.20	−0.63

All cash amounts are millions. At 60%, the alternative equity payoff is 3.20.

Higher rates eliminate marginal projects



Each project has a threshold. A collection of projects creates an investment response.

IS means Investment–Saving

IS lists the output/rate pairs where
planned spending equals production.

In a closed-economy equilibrium,
planned investment equals planned national saving.

Why IS slopes downward

A higher rate makes fewer investments worthwhile.

Lower investment means less planned expenditure.

Equilibrium output is lower.

Output is on the horizontal axis. The rate is on the vertical axis.

Households respond to disposable income

$$C = C_0 + c(Y - T), \qquad 0 < c < 1.$$

Symbol	Meaning
C	Planned consumption
C_0	Consumption independent of current income
T	Taxes less transfers
c	Extra consumption per extra unit of disposable income

Investment responds to the rate

$$I = I_0 - bi, \qquad b > 0.$$

Symbol	Meaning
I	Planned investment
I_0	Spending independent of the current rate
b	Response to the nominal rate, with other inputs fixed

Fixed expected inflation and credit spreads make i a proxy for the relevant real financing cost.

Combining spending gives the IS equation

G: government consumption, fixed in this experiment.

$$Y = C + I + G.$$

$$\boxed{Y = \frac{C_0 - cT + I_0 + G - bi}{1 - c}.}$$

The equation records all goods-market equilibria for the chosen assumptions.

LM means Liquidity preference–Money supply

LM lists output/rate pairs where
desired money balances equal money supply.

Holding money is useful for payments.
Holding it also gives up interest on other assets.

Why classic LM slopes upward

$$\frac{M}{P} = L(Y, i), \qquad \frac{di}{dY} = -\frac{L_Y}{L_i} > 0.$$

More output means more payments and a greater desire to hold money.

With M/P fixed, i rises until the opportunity cost of cash offsets that extra demand.

The central bank chooses M. Private money demand and the bond market determine the clearing i.

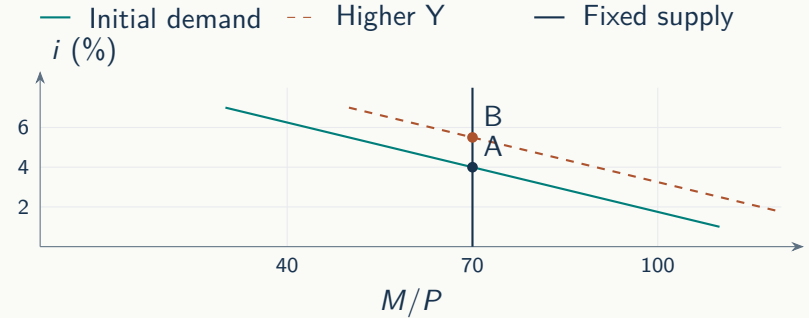
Agents adjust portfolios, and bond prices move

More transactions raise cash demand → Investors try to sell bonds for cash → Bond prices fall and yields rise

The total money stock is unchanged. Higher i reduces desired cash holdings.

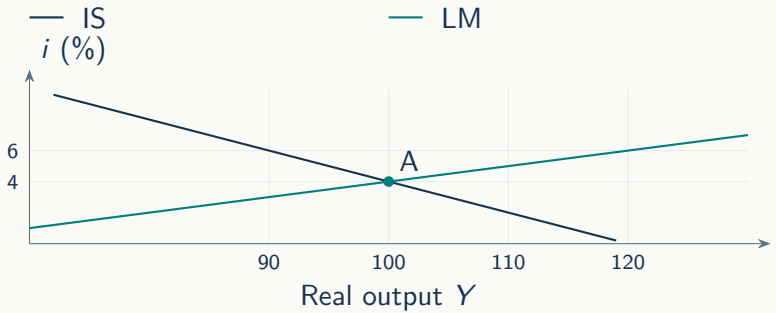
It is a portfolio opportunity-cost mechanism. Spending need not disappear to clear the money market.

Higher output shifts money demand to the right



At the same fixed M/P, higher real activity raises the clearing market rate.

The intersection balances both markets



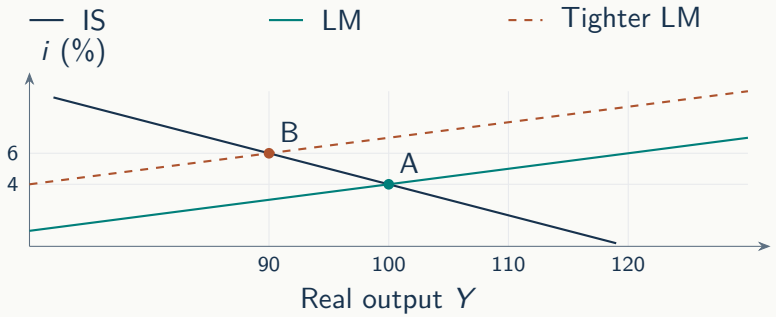
At A, planned expenditure equals output and money demand equals supply.

A movement along a curve versus a shift

Movement along IS	**A shift of IS**
A change in the rate, holding other spending inputs fixed.	A change in spending at each given rate.

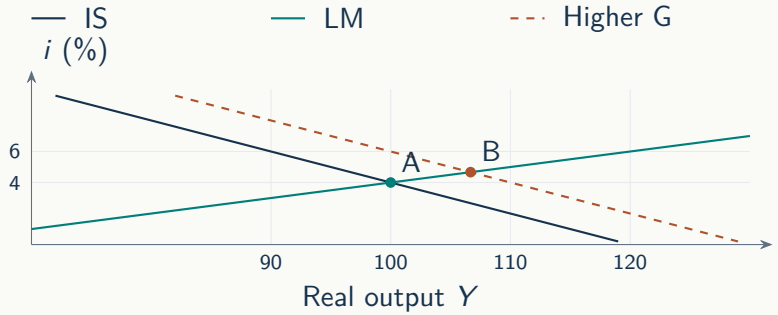
First identify what changed. Then identify which relationship changes.

Monetary tightening shifts classic LM



Lower real money supply: A becomes B. The rate rises and output falls.

Fiscal expansion with an unchanged money supply



The same IS shift raises output and the rate. Private investment partly falls.

A demand shock versus a supply shock

Lower confidence shifts planned spending inward.

Lower productive capacity moves potential output.

Policy faces different inflation consequences in the two cases.

LECTURE 2

Inflation and policy feedback

Why can a central bank raise nominal rates by more than inflation?

The neutral real rate is a benchmark

r^*: the real rate consistent with output at potential and stable inflation.

The actual real rate may be above or below it.

Neutral is an estimate. It depends on the economy's resources, preferences and opportunities.

At the target, a neutral nominal benchmark

$$\bar{i} \approx r^* + \pi^*.$$

π^*: the inflation target.

$$r^* = 1\%, \quad \pi^* = 2\% \quad \Rightarrow \quad \bar{i} \approx 3\%.$$

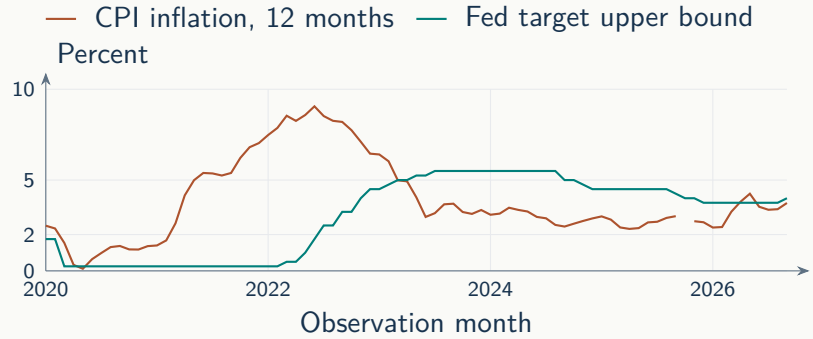
If inflation rises, the nominal rate must catch up

Suppose actual and expected inflation rise from 2% to 3%.

$$\text{To keep a 1\% real rate:} \quad \bar{i} = 4\%.$$

To add restraint, an illustrative rule raises it further to 4.5%.

US inflation and policy after Covid



Jan 2020–Sep 2026. BLS/Federal Reserve via FRED. Policy: month-end. Missing Oct 2025 CPI is a gap.

The output gap measures pressure on resources

$$x_t = 100 \frac{Y_t - Y_t^*}{Y_t^*}.$$

Symbol	Meaning
Y_t	Actual real output
Y_t^*	Potential real output
x_t	Output gap in percent

Positive gap: output above potential. Negative gap: slack.

Weak activity changes the policy trade-off

Keep inflation at 3%, but put output 2% below potential.

$$\text{Illustrative response:} \quad 4.5 - 0.5 \times 2 = 3.5\%.$$

Inflation calls for restraint. Slack calls for support.

A central banker might follow a rule like this

$$\bar{i}_t = r^* + \pi_t + 0.5(\pi_t - \pi^*) + 0.5x_t.$$

One possible policy reaction: respond to inflation and the output gap.

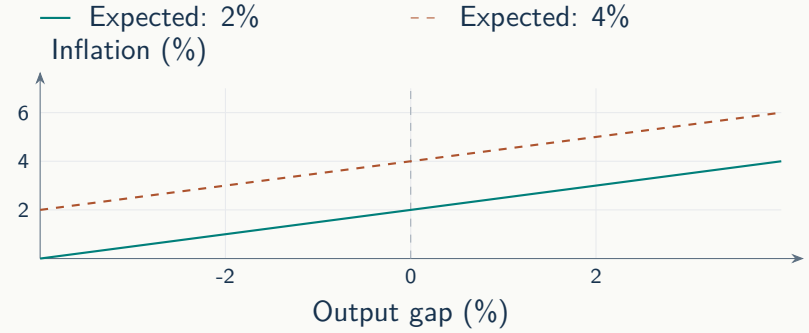
Illustrative Taylor-style rule. Rates and gaps use percentage points; this is not a commitment.

A simple Phillips relationship

$$\pi_t = \pi_t^e + \kappa x_t + u_t.$$

Symbol	Meaning
π_t^e	Expected inflation for this period
x_t	Output gap, as defined earlier
$\kappa > 0$	Inflation response to demand pressure
u_t	Cost or supply disturbance

Expectations shift the Phillips curve



Hypothetical $\kappa = 0.5$ and no supply disturbance.

Inflation deviations are easier to analyse

$$d_t = \pi_t - \pi^*.$$

Symbol	Meaning
d_t	Inflation deviation from target, in percentage points
π_t	Inflation rate
π^*	Inflation target

Inflation of 3% with a 2% target means $d_t = 1$.

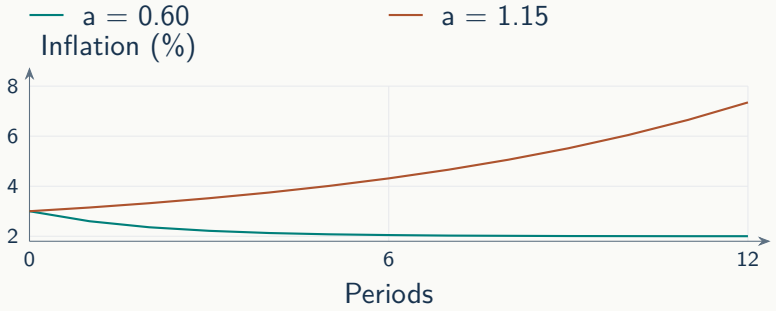
A simple model of inflation persistence

$$d_t = a\, d_{t-1} + u_t.$$

Symbol	Meaning
a	Net carry-over from the previous inflation deviation
u_t	A new inflation disturbance this period

After an initial shock, set later disturbances to zero and compare values of a.

The same shock can decay or amplify



Target 2%, initial deviation 1 point. Stable when $|a| < 1$. Original hypothetical paths.

Exercise: inflation rises while output falls

An energy shortage raises costs and reduces productive capacity.

Compare holding the nominal rate fixed with raising it.

Which assumptions determine the change in the real rate?

LECTURE 2

Modern rate targeting

What changes when the central bank chooses the rate?

Choose the rate, and let the monetary quantity adjust

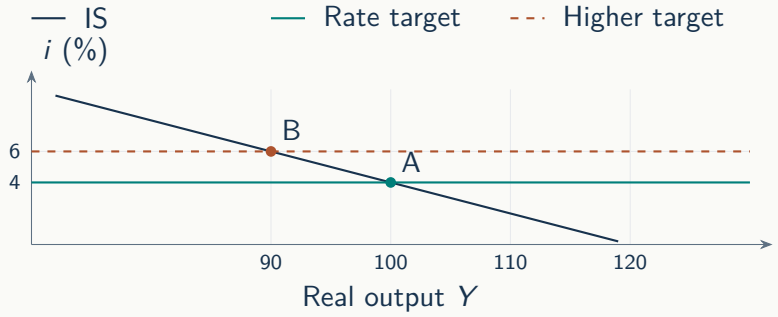
$$i = \bar{i}, \qquad \frac{M}{P} = L(Y, \bar{i}).$$

The central bank chooses the target. It accommodates the quantity needed to implement it.

For this experiment, the target is fixed as Y changes.

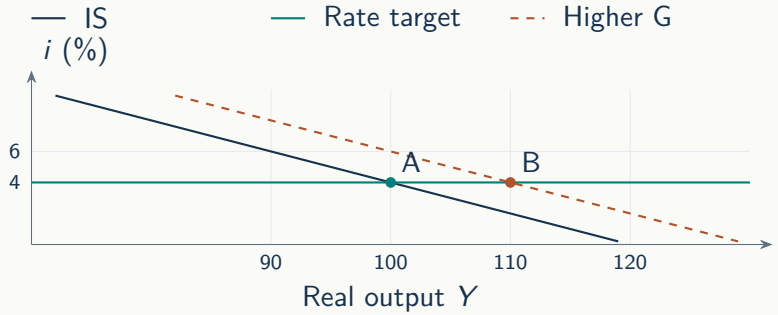
Across meetings, the chosen target can respond to inflation and activity.

With a rate target, policy sets a horizontal line



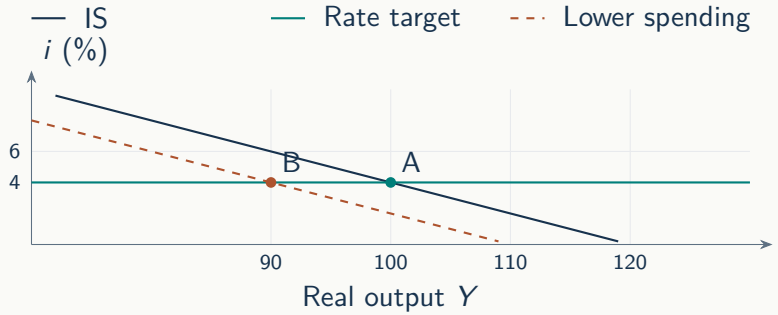
Chosen $i = \bar{i}$ rises from 4% to 6%; output falls from 100 to 90.

Fiscal expansion with an unchanged rate target



Higher government spending shifts IS outward. At a fixed rate, output rises.

A private-demand slowdown shifts IS inward



Lower autonomous consumption or investment reduces output at a fixed target.

LECTURE 2

Optional mathematical extension

How does policy change an inflation recurrence?

Optional extension: what changes the feedback?

We will connect three mechanisms:

Inflation pressure, the demand response to real rates, and a policy response to inflation.

Each relationship comes next on its own slide. The coefficients describe this toy model only.

Inflation responds to activity and past inflation

$$d_t = \alpha d_{t-1} + \kappa x_t + u_t.$$

New coefficient	Meaning
α	Carry-over of the previous inflation deviation
$\kappa > 0$	Inflation response to a one-point output gap

d is the inflation deviation, x the output gap, and u a new disturbance.

Real-rate restraint reduces the output gap

$$x_t = -B(r_t - r^*), \qquad B > 0.$$

B: how strongly demand responds to the real-rate gap.

Higher real rates relative to neutral reduce demand.
B is an aggregate sensitivity, distinct from investment's earlier coefficient b.

Policy adds real restraint when inflation exceeds target

$$r_t - r^* = \phi d_{t-1}, \qquad \phi > 0.$$

ϕ: the real-rate response to a one-point inflation deviation.

The nominal policy rate must also offset expected inflation to deliver this real response.

Substitution gives a stability condition

$$d_t = \underbrace{(\alpha - \kappa B \phi)}_{a} \, d_{t-1} + u_t.$$

$$\boxed{|\alpha - \kappa B \phi| < 1.}$$

If $\alpha = 1.15$ and $\kappa B \phi = 0.55$, then $a = 0.60$.

Policy supplies negative feedback in this specified model.

A two-date endowment economy

$$U = \ln c_0 + \beta \ln c_1.$$

c_0, c_1: consumption now and later. $\beta > 0$: patience.

e_0, e_1: the economy's available resources on the two dates.

U: lifetime utility. The real bond transfers purchasing power between dates.

The individual intertemporal budget

$$c_0 + \frac{c_1}{1+r} = e_0 + \frac{e_1}{1+r}.$$

Saving one unit now buys a claim to future resources.

The individual chooses consumption given the rate.

Market clearing pins down a real rate

$$\frac{1}{c_0} = \beta(1+r)\frac{1}{c_1}, \qquad c_0 = e_0, \quad c_1 = e_1.$$

$$\boxed{1 + r = \frac{e_1}{\beta e_0}.}$$

More resources now lower the rate, other inputs fixed.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.